

Code-Layout, Readability and Reusability

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Team ID	6ab7f3e4b8 bf817b7f72 4e1d
Project Name	FitBuddy
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Standard 4-space Python PEP 8 indentation applied consistently.
2	Proper File Structure	Files and folders are logically organized	Yes	Clean separation between app routes, database models, templates, and static assets.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive naming conventions used across Pydantic schemas and route parameters
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly indicate operations (e.g., plan generation, feedback handling)
5	Code Comments	Inline and block comments explain logic	Yes	Key logic blocks and AI integration steps are properly documented
6	Modular Design	Code is split into reusable functions/modules	Yes	Isolated modules for database logic, Gemini generators, and routes.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Clean codebase with minimal boilerplate.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation via Pydantic and safe exception handling for API calls.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Database Session & Models (database.py)	Python / SQLAlchemy	Reused across all FastAPI route handlers for data persistence	High

2	Gemini AI Generators (gemini_generator.py)	Python / Google GenAI SDK	Reused for core workout generation and update routes	High
3	Base Templates & Layouts	HTML / Jinja2	Reused across index, result, and admin dashboard views	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean, modular separation of concerns across dedicated files
Readability	5	Intuitive variable naming, clean syntax, and adherence to Python standards.
Reusability	5	Core database and AI logic are modularized for easy extension.
Documentation / Comments	5	Thorough setup instructions and clear inline notes provided.
Overall Score	5	